INCH-POUND

A-A-52139
November 30, 1993

COMMERCIAL ITEM DESCRIPTION

ICE MAKING MACHINE, CUBE

The General Services Administration has authorized the use of this commercial item description in preference to Military Specification MIL-I-11867 (see 5.5).

1. ABSTRACT

1.1 Classification. This commercial item description covers two types, eleven sizes, and two grades of cube ice making machines as follows (see 5.2).

Type I - Air cooled condenser
Type II - Water cooled condenser

Size 1 - 50 pound capacity
Size 2 - 100 pound capacity
Size 3 - 150 pound capacity
Size 4 - 200 pound capacity
Size 5 - 300 pound capacity
Size 6 - 400 pound capacity
Size 7 - 500 pound capacity
Size 8 - 600 pound capacity
Size 9 - 800 pound capacity
Size 10 - 900 pound capacity
Size 11 - 1200 pound capacity

Grade A - Carbon steel cabinet or stainless steel cabinet (300 series or 400 series)
Grade B - Stainless steel cabinet (300 series only)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4110

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1.2 CID based part identification number (PIN). A document based PIN to identify types, sizes, and grades of items is included in section 5. This identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor (see 5.6).

2. SALIENT CHARACTERISTICS

2.1 Design and construction. The ice making machine shall be a complete self-contained unit composed on an ice making section and a storage bin. The machine shall have a means of harvesting and storing ice. The ice making machines shall be provided with a water circulating system, motor or motor drives, and all necessary piping, wiring, controls, safety devices, and any other components or accessories necessary to provide an automatic operating ice making machine with automatic control of ice level in the storage bin. The ice making machine's maximum dimensions shall be as shown in table I. The maximum height dimensions include the ice making machine being mounted on 6 inch adjustable legs.

TABLE I. Maximum overall dimensions (inches)

Size	Length	Depth	Height
1	18	24	36
2	32	27	45
3	36	38	72
4	36	38	72
5	49	34	73
6	49	38	96
7	49	38	96
8	53	38	96
9	45	35	96
10	73	35	100
11	53	35	98

2.1.1 Ice making section. The ice making section (all components except the ice storage bin) shall be totally enclosed in a structurally sound housing with removable access panels for repair, replacement, and lubrication of all internal components. Access panels shall be capable of being detached and re-attached without the use of special tools. Contact areas between dissimilar metals shall be insulated to minimize galvanic corrosion.

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2.1.2 Storage bin. The storage bin shall be an integral part of the ice making machine. The storage bin liner shall be fabricated from plastic, or from 24 gauge (0.0239 inch) minimum stainless steel 300 series conforming to ASTM A 167, or 400 series conforming to ASTM A 176. The bin liner shall not leak. Insulation between the exterior and interior of the storage bin shall be of sufficient rating, density, and amount so that no condensation will form on the exterior of the storage bin when ice is stored in the bin. The insulation shall not be of the loose-fill type, shall not be combustible, shall not support combustion, and shall not release toxic fumes when exposed to a high temperature. The ice storage bin shall have a drainage opening. Unless otherwise specified (see 5.2), the ice storage bin shall be sized to contain the amount of ice listed below for the given sizes of ice making machines:

Size 3 - 150 pound production - 300 pound storage bin
Size 6 - 400 pound production - 500 pound storage bin
Size 9 - 800 pound production - 800 pound storage bin

For sizes of ice making machines not listed above, the ice storage bin shall be sized to contain the amount of ice produced by the ice making section during a 12-hour period, unless otherwise specified (see 5.2).

2.2 Capacity. The ice making machine shall have the capacity to make the amount of ice listed in table II, under the conditions specified therein. Ice produced shall be cube ice. Cube ice normally refers to a fairly uniform product that is hard, solid, usually (but not necessarily) clear, and weighs less than 2 ounces per piece, as distinguished from flaked, crushed, or fragmented ice. The ice may be formed by any method and in any shape consistent with the manufacturer's stndard product.

TABLE II. Capacity conditions

Type	Size	Ambient air temp (degrees F)	Potable water supply temp (degrees F)	Condenser inlet water temp (degrees F)	Minimum capaacity for 60Hz machine (lbs/24 hrs)	Minimum capacity for 50Hz machine (lbs/24 hrs)
I	1	90	70	None	50	42
I	2	90	70	None	100	83
I	3	90	70	None	150	125
I	4	90	70	None	200	167
I	5	90	70	None	300	250
I	6	90	70	None	400	333
I	7	90	70	None	500	417
I	8	90	70	None	600	500
I	9	90	70	None	800	667
I	10	90	70	None	900	750
I	11	90	70	None	1200	1000

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TABLE II. Capacity conditions (cont'd)

Type	Size	Ambient air temp (degrees F)	Potable water supply temp (degrees F)	Condenser inlet water temp (degrees F)	Minimum capacity for 60Hz machine (lbs/24 hrs)	Minimum capacity for 50Hz machine (lbs/24 hrs)
II	1	90	70	70	50	42
II	2	90	70	70	100	83
II	3	90	70	70	150	125
II	4	90	70	70	200	167
II	5	90	70	70	300	250
II	6	90	70	70	400	333
II	7	90	70	70	500	417
II	8	90	70	70	600	500
II	9	90	70	70	800	667
II	10	90	70	70	900	750
II	11	90	70	70	1200	1000

2.3 Type I ice making machine. The type I ice making machine shall be provided with an air cooled condenser.

2.4 Type II ice making machine. The type II ice making machine shall be provided with a water cooled condenser.

2.5 Grade A ice making machine. The exterior of the grade A ice making machine shall be fabricated of low carbon steel sheet conforming to ASTM A 366 or from 300 series stainless steel conforming to ASTM A 167 or 400 series stainless steel conforming to ASTM A 176 with a minimum thickness of 22 gauge (0.0299 inch). The exterior of the ice making machine fabricated of the grade A low carbon steel shall have the manufacturer's standard commercial finish, provided it conforms to NSF Standard No. 12.

2.6 Grade B ice making machine. The exterior of the grade B ice making machine shall be fabricated of 300 series stainless steel conforming to ASTM A 167 with a minimum thickness of 20 gauge (0.0355 inch).

2.7 Refrigeration unit. The refrigeration unit shall be an integral part of the complete ice making machine. The refrigeration unit shall be completely self-contained, pre-charged, and wired. The compressor and compressor motor shall be of the type and size to obtain the capacities listed in table II.

2.7.1 Refrigerant. The system refrigerant shall be provided by the manufacturer and shall be that which is acceptable to the Environmental Protection Agency (EPA) at the time of manufacture. The compressor lubrication oil shall be compatible with the refrigerant used. The system shall be labeled for refrigerant and lubrication oil as provided.

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2.8 Electrical requirements. Ice making machines with compressor motors up to and including 3/4 horsepower shall operate on a 120 volt, 60 Hz, 1-phase electrical system. Ice making machines with compressor motors over 3/4 horsepower shall operate on a 208/230 volt, 60 Hz, 1-phase electrical system. When 50 Hz frequency is specified (see 5.2), the contracting officer shall provide the electrical requirements. The motors shall comply with National Electrical Manufacturers Association (NEMA) Standard MG 1.

2.8.1 Cord and attachment plug. Ice making machines shall be equipped with a 6 to 10 foot long UL listed, three-conductor cord with a NEMA approved plug, or shall have provisions for a permanent electrical connection (see 5.2).

2.9 Standards of compliance. The ice making machines shall comply with the applicable requirements of National Fire Protection Association (NFPA) Standard No. 70, National Sanitation Foundation (NSF) Standard No. 12, Underwriters Laboratories, Inc. (UL) Standard No. 563, the American Society for Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) Standard No. 29, and with the Air-Conditioning and Refrigeration Institute (ARI) Standard No. 810 (see 3.2).

2.10 Ice crushing mechanism. When specified, an automatic ice crushing mechanism shall be furnished with the ice making machine (see 5.2).

2.11 Data name plate (Government unique requirement (see 5.4)). The ice making machine shall be furnished with a data name plate. The data name plate shall be made of minimum 20 gauge (0.0355 inch) corrosion-resisting metal and attached to the ice making machine by rivets, screws, or by soldering in such a manner as to meet the applicable requirements for this equipment. The plate shall contain the following information which shall be stamped, engraved, or applied by photosensitive means:

National Stock Number
Procurement Contract Number
Specification Data
Manufacturer's Name, Address, and Telephone Number
Supplier's Name, Address, and Telephone Number (List only if different from Manufacturer)
Manufacturer's Model Number
Government Approved Manual Number (when specified see 5.2)

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the ice making machine. Lettering shall not be less than 1/16 inch high.

2.12 Instruction plate. Instruction plates made of corrosion-resisting metal shall be attached to the ice making machine. The instruction plates shall have the wiring diagram, refrigeration cycle diagram, instructions

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for start-up, operation, shutdown and preventative maintenance procedures, and shall identify the type of refrigerant supplied in the system. In addition, information as required by UL Standard No. 563 shall also appear on the plate. Lettering shall not be less than 1/16 inch high.

2.13 Service and maintenance. All major assemblies and installed components shall be accessible for maintenance, repair, service, and replacement with minimum disturbance to adjacent components in the ice making machine and without usage of special tools.

2.14 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this commercial item description unless otherwise specified.

2.15 Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

2.16 Workmanship. All components and assemblies of the ice making machines shall be free from dirt and other harmful extraneous material. Ice making machines shall also be free of burrs, slivers, rough die, tool and grind marks, dents, and cracks.

3. QUALITY ASSURANCE

3.1 Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and except for any Government unique requirements specified, conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial market place, or is the same product that has successfully been delivered

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to the government on a previous contract or purchase order. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 Compliance. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the described item conforms to the requirements of National Fire Protection Association (NFPA) Standard No. 70, National Sanitation Foundation (NSF) Standard No. 12, Underwriters Laboratories, Inc. (UL) Standard No. 563, the American Society for Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) Standard No. 29, and with the Air Conditioning and Refrigeration Institute (ARI) Standard No. 810 (see 2.9).

3.2.1 National Fire Protection Association (NFPA). The contractor shall submit to the contracting officer a written certificate of compliance as proof that the ice making machine meets the applicable requirements of NFPA Standard No. 70.

3.2.2 National Sanitation Foundation (NSF). Acceptable evidence of meeting the requirements of NSF Standard No. 12 shall be a listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished ice making machine.

3.2.3 Underwriters' Laboratories, Inc. (UL). Acceptable evidence of meeting the applicable requirements of the Underwriters' Laboratories Standard No. 563 shall be the UL label, a UL listing mark, or a certified test report from a recognized independent laboratory, acceptable to the Government, indicating that the ice making machine conforms to the applicable requirements of the specified UL standard.

3.2.4 American Society for Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE). The contractor shall submit to the contracting officer a written certificate of compliance as proof that the ice making machine meets the applicable requirements of ASHRAE Standard No. 29.

3.2.5 Air Conditioning and Refrigeration Institute (ARI). The contractor shall submit to the contracting officer a written certificate of compliance as proof that the ice making machine meets the applicable requirements of ARI Standard No. 810.

4. PACKAGING

4.1 Packaging and packing. Packaging and packing shall be commercial or export as specified (see 5.2).

4.1.1 Commercial packaging and packing. Each ice making machine shall be packaged and packed in accordance with the level C packaging and packing requirements of MIL-R-3593.

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4.1.2 Export packaging and packing. Each ice making machine shall be packaged and packed in accordance with the level A packaging and packing requirements of MIL-R-3593.

4.2 Marking (commercial and export). Marking of unit packs and shipping containers shall be in accordance with MIL-STD-129.

5. NOTES

5.1 Intended use. The ice making machine is intended to be used in dining facilities, hospitals, and other military installations.

5.2 Acquisition Requirements. Acquisition documents must specify the following:

a. Title, number and date of this document.
b. Type, size, and grade of ice making machine required (see 1.1).
c. When ice storage bin capacity is other than specified (see 2.1.2)
d. When electrical characteristics other than specified are required, the contracting officer shall provide voltage, phase, cycle, and other necessary requirements to facilitate operation and installation (see 2.8).
e. Type of electrical connection required (see 2.8.1).
f. When an ice crushing mechanism is required (see 2.10).
g. The government approved manual number is to be included on the data name plate when manuals are purchased under the contract (see 2.11).
h. Selection of the applicable levels of packaging and packing required (see 4.1).

5.3 Source of documents.

5.3.1 Sources of governmental documents. Copies of military and federal documents are available from:

Standardization Documents Order Desk
Building 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

5.3.2 Sources for nongovernment association documents.

ARI Standard No. 810 - Automatic Commercial Ice-Makers, is available from:

Air-Conditioning and Refrigeration Institute (ARI)
1501 Wilson Boulevard
Arlington, VA 22209

ASHRAE Standard No. 29 - Methods of Testing Automatic Ice Makers, is available from:

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American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)
1791 Tullie Circle, NE
Atlanta, GA 30329

ASTM A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip, and
ASTM A 176 - Stainless and Heat Resisting Chromium Steel Plate, Sheet and Strip, and
ASTM A 366 - Steel, Sheet, Carbon, Cold-Rolled, Commercial Quality, are available from:

The American Society for Testing and Materials (ASTM)
1916 Race Street
Philadelphia, PA 19103-1187

NEMA Standard MG 1 - Motors and Generators, is available from:

National Electrical Manufacturers Association (NEMA)
2101 L Street, N.W.
Suite 300
Washington, DC 20037

NFPA Standard No. 70 - National Electrical Code, is available from:

National Fire Protection Association (NFPA)
1 Batterymarch Park
Quincy, MA 02269

NSF Standard No. 12 - Automatic Ice Making Equipment, is available from:

National Sanitation Foundation
3475 Plymouth Road
P.O. Box 1468
Ann Arbor, MI 48106

UL Standard No. 563 - Ice Makers, is available from:

Underwriters Laboratories, Inc. (UL)
333 Pfingsten Road
Northbrook, IL 60062

5.4 Government unique requirements. Whenever "(Government unique requirements)" is included in the title of a paragraph under "salient characteristics", it is meant that the requirement is something that is not normally offered to the commercial market place by the manufacturer.

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5.5 Supersession data. The types, sizes, and grades of cube ice making machines covered by this commerciel item description replace the items covered by MIL-I-11867 as indicated below:

CID A-A-52139	MIL-I-11867
Type I - Air cooled condenser	Type I - Water cooled condenser
Type II - Water cooled condenser	Type II - Air cooled condenser
Size 1 - 50 pound capacity	Size 9 - 50 pound capacity
Size 2 - 100 pound capacity	Size 10 - 100 pound capacity
Size 3 - 150 pound capacity	Size 11 - 150 pound capacity
Size 4 - 200 pound capacity	Size 1 - 200 pound capacity
Size 5 - 300 pound capacity	Size 2 - 300 pound capacity
Size 6 - 400 pound capacity	Size 3 - 400 pound capacity
Size 7 - 500 pound capacity	Size 4 - 500 pound capacity
Size 8 - 600 pound capacity	Size 5 - 600 pound capacity
Size 9 - 800 pound capacity	Size 7 - 800 pound capacity
Size 10 - 900 pound capacity	Size 6 - 900 pound capacity
Size 11 - 1200 pound capacity	Size 8 - 1200 pound capacity
Grade A - Carbon steel cabinet or stainless steel cabinet (300 series or 400 series)	Grade A - Same
Grade B - Stainless steel cabinet (300 series only)	Grade B - Same

5.6 Part identification number (PIN). The PINs to be used for cube ice making machines acquired by this CID are created as follows:

A 52139 -I -1 -A (Example of a PIN: A52139-I-1-A)

- -A: Grade
- -1: Size
- -I: Type
- 52139: CID number
- A: Designates a Commercial Item Description

MILITARY INTERESTS:

Custodians
Army - GL
Navy - YD
Air Force - 99

Review Activities
Army - MD, QM
Navy - MC
Air Force - 35, 82
DLA - GS

User Activity
Navy - SA

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Army - GL

(Project 4110-0472)

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-52139	93-11-30

3. DOCUMENT TITLE
Ice Making Machine, Cube

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-5175	(2) AUTOVON/DSN 256-5175
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-WEE Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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